

BANKING DASHBOARD PROJECT

Risk Analytics in Banking & Financial Services

1. Problem Statement

Develop a basic understanding of **risk analytics** in banking and financial services and understand how data is used to minimise the risk of losing money while lending to customers.

Banks face significant challenges in assessing the creditworthiness of loan applicants. Incorrect lending decisions can lead to high default rates, non-performing assets (NPAs), and substantial financial losses. There is a need for data-driven tools that help decision-makers evaluate applicant profiles effectively and approve or reject loans based on quantifiable risk indicators.

2. Objectives

- Build interactive Power BI dashboards that present key banking metrics and KPIs in a clear, actionable format.
- Enable decision-makers to analyse loan and deposit portfolios by gender, income band, nationality, occupation, and banking relationship type.
- Identify patterns that help minimise lending risk (e.g., clients likely to repay vs. higher-risk profiles).
- Provide a consolidated view of Total Clients, Total Loan, Total Deposit, Business Lending, and related account balances.

3. About the Dataset

The dataset contains information about bank details and various client details. It consists of multiple tables that are interlinked through primary and foreign keys.

Tables included:

- Banking Relationship
- Client-Banking
- Gender
- Investment Advisor
- Period

These tables enable analysis across dimensions such as client demographics, account types, loan products, deposits, and advisor relationships.

4. Data Cleaning & Feature Engineering

The following transformations were performed on the Client-Banking table to support richer analysis:

- **Engagement Timeframe:** New column indicating the timeline of clients in the bank.

- **Engagement Days:** Number of days the client has spent with the bank since the joining date (calculated using DATEDIFF).
- **Income Band:** Binned Estimated Income: Low (< 100,000), Mid (< 300,000), and High (remaining).
- **Processing Fees:** Derived from Fee Structure – if fee structure is High, Processing Fee = 0.05 (and similarly for other bands).

5. Scope of the Project

This project focuses on developing Power BI dashboards for **Loan Analysis**, **Deposit Analysis**, and a **Summary / Home** view. The dashboards support filtering by Joining Year, Gender, Banking Relationship, and Institution Advisor, allowing stakeholders to drill into risk and performance metrics.

6. Expected Outcome

A set of interactive dashboards that empower the bank to make informed lending decisions based on applicant profiles, reduce the risk of monetary loss, and gain insights into client distribution, loan exposure, and deposit composition across key demographic and relationship segments.